

SE LAB 5 REPORT:

TABLE:

Issue	Type	Severity	Line(s) (Original)	Description	Fix Approach
Architectural Flaw	Convention (W0603)	Medium/Low	N/A (Global Structure)	Program relied on global state (stock_data), flagged by Pylint for poor maintainability.	Major Refactoring: Encapsulated state and functions within the Inventory class , eliminating the need for the global keyword.
Dangerous eval() usage	Security (Bandit B307)	High	78	The use of eval() allowed the execution of arbitrary code ¹ , posing a critical security risk.	Removed eval() entirely from the main() function.
Mutable Default Argument	Bug (Pylint W0102)	High	12	The default argument logs=[] was a list shared across all calls, causing data corruption ² .	Resolved by Class Refactor: State (self.inventory_logs) is now managed per instance.

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Catching Broad Exception	Bug (Pylint W0703)	Medium	24	The bare except: clause in removeitem() masked all errors, including system-exiting ones ³ .	Replaced with specific exceptions: KeyError and ValueError .
Unsafe File Access	Bug (Pylint W1514)	41, 46	Manual open/close risked resource leaks.	Implemented the safe with open(...) as file_handler: context manager.	
Input Validation	Robustness	Medium	15, 27	Functions lacked explicit checks for numeric qty and valid item type.	Added try/except checks for quantity conversion in add_item and remove_item.
Naming Convention	Style (Pylint C0103)	Low	15, 27, 34, etc.	Function names (e.g., addItem, getQty) did not follow Python's snake_case .	Renamed all methods (e.g., add_item, get_qty) to be snake_case.

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				standard	
Iteration Style	Style (Pylint C0200)	Low	53, 58	Used indirect iteration: for i in stock_data :.	Used explicit iteration: for item, qty in self.stock_data.items():.
Formatting/Docstrings	Style (Pylint C0114/C0305)	Low	N/A	Missing module docstring , and use of old % string formatting.	Added module docstring and converted logging/output to f-strings where appropriate ⁴ .
Flake8 Compliance	Style (Flake8 E501)	Low	Multiple	Many lines exceeded the 79-character limit ⁵ .	Applied strict line continuation and breaks to comply with PEP 8 standards.